DECI project : `Ford gobike`

exploratory data analysis

The Ford GoBike System Data contains information about a bike-sharing service that was available in the San Francisco Bay Area. It includes details about bike trips, such as where and when they started and ended, how long they lasted, and the type of users who rode them. This data can help people understand how bike-sharing is used in the city, including travel patterns and popular routes.

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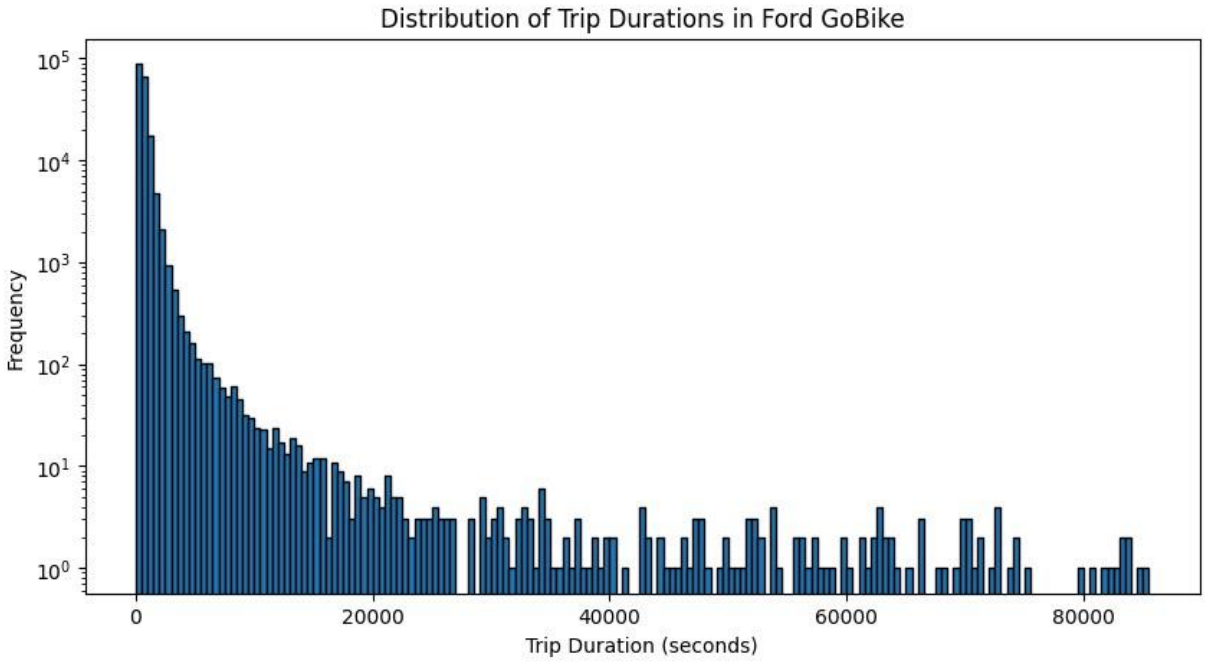
Assesment Report

1. there are unnecessary column such as start_station_id, start_station_latitude, start_station_longitude ,end_station_id, end_station_latitude and end_station_longitude
2. data type of the columns 'start_time' and 'end_time' should be datetime
3. nan values of gender column should be replaced by other
4. add 'year-month' column for better analysis
5. there is nan values in 'memeber_birth_year'
6. add 'age' column and 'age_group' column for better analysis

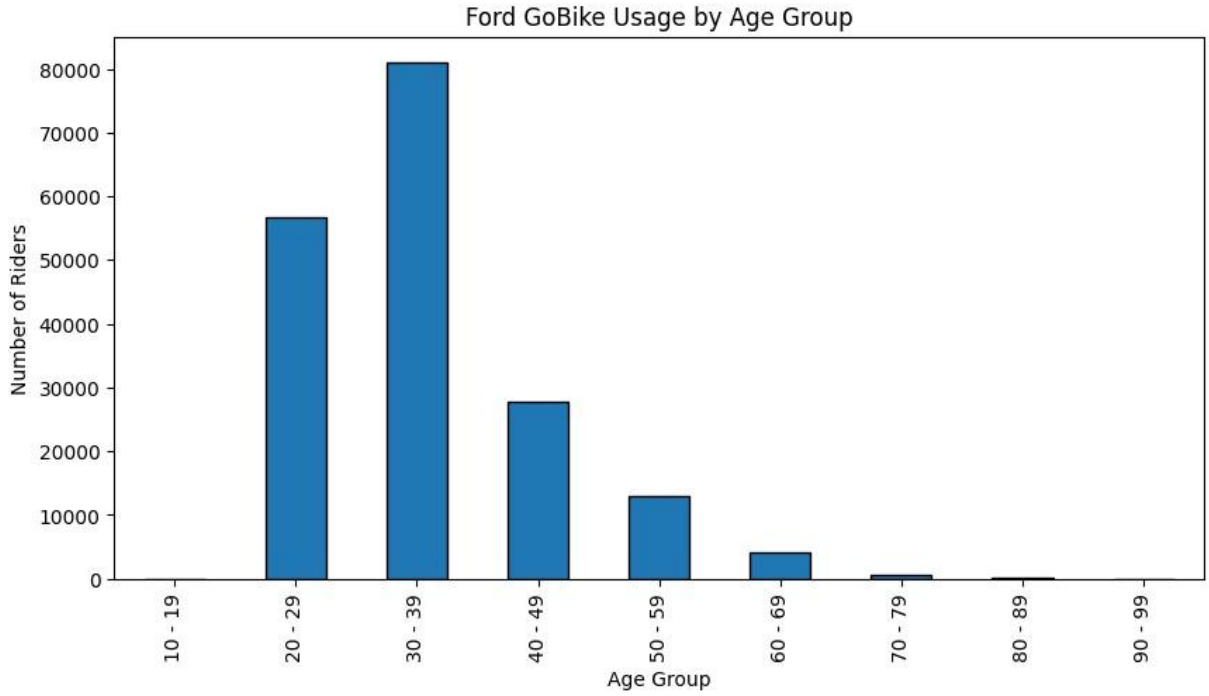
Univariate Exploration

What is the distribution of trip durations in the Ford GoBike dataset?



In this analysis, we look at the distribution of trip durations in the Ford GoBike dataset using a histogram. Since the data is heavily right-skewed due to extremely long trips, we apply a logarithmic scale on the y-axis to better visualize the data. Even with this adjustment, we see that most trips are fairly short, with the majority lasting under 1,000 seconds (~16 minutes). However, there are some very long trips (over 80,000 seconds or ~22 hours), which may suggest outliers, system errors, or potential misuse. This indicates that most users take quick rides, but further analysis is needed to investigate outliers and usage patterns across different rider types

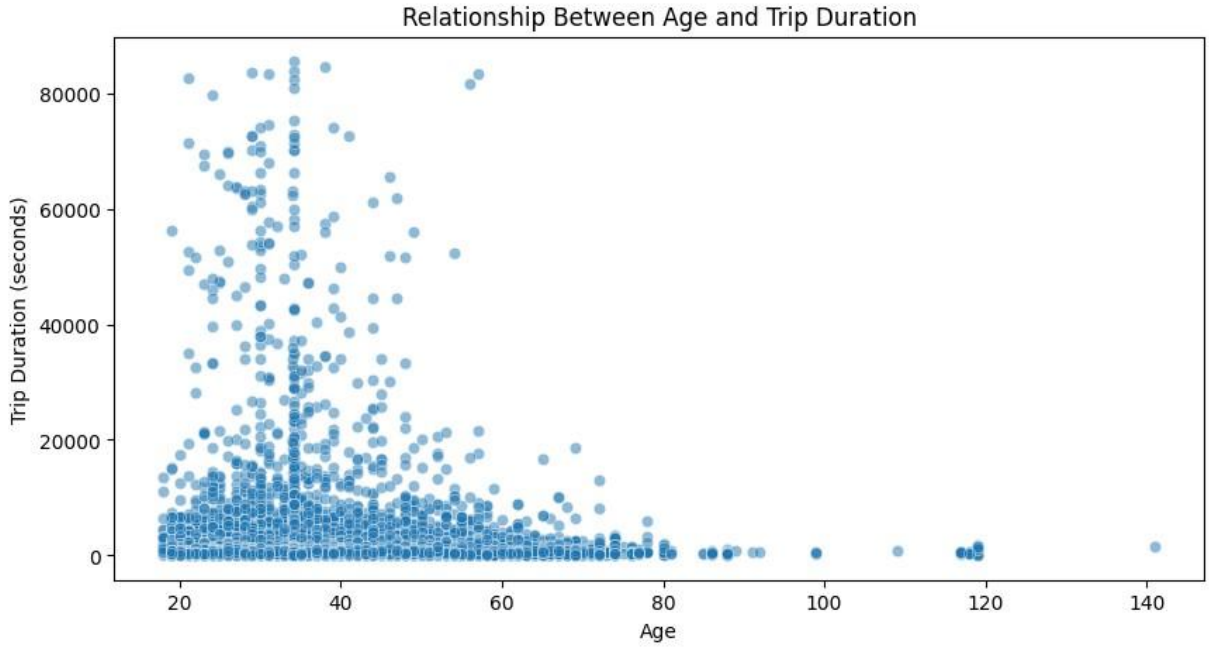
What age group uses bikes the most?



In this analysis, we examine bike usage across different age groups to identify which group rides the most. The bar chart shows that the 20-39 age range dominates bike usage, with the 30-39 age group having the highest number of riders, followed closely by 20-29-year-olds. After age 40, there is a gradual decline in usage, with significantly fewer riders in the 50+ age range. The drop-off is sharp after 60, suggesting that bike-sharing is primarily popular among young and middle-aged adults, likely due to commuting and convenience.

Bivariate Exploration

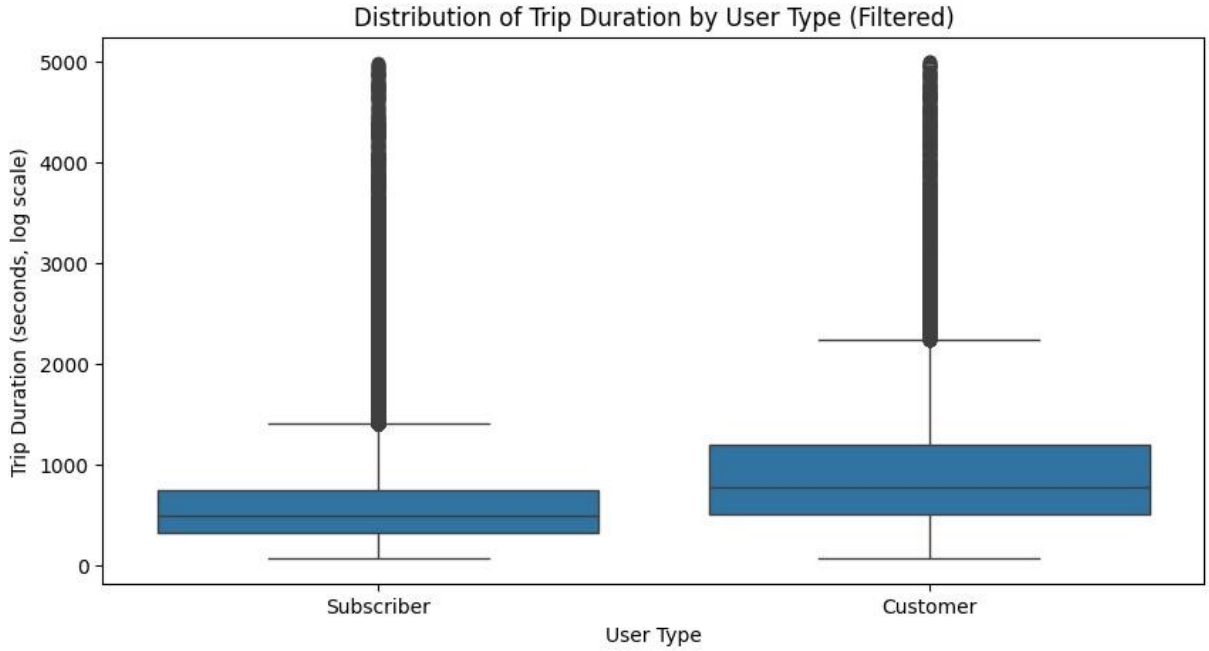
How does trip duration vary with age?



In this analysis, we explore the relationship between age and trip duration to see if certain age groups tend to take longer rides. The scatter plot shows that most trips are short, clustering under 5,000 seconds (~83 minutes). However, younger riders, particularly those in their 20s and 30s, have a higher likelihood of taking longer trips, with some exceeding 20,000 seconds (~5.5 hours). As age increases, trip durations become shorter and more consistent, suggesting that older riders prefer quicker trips.

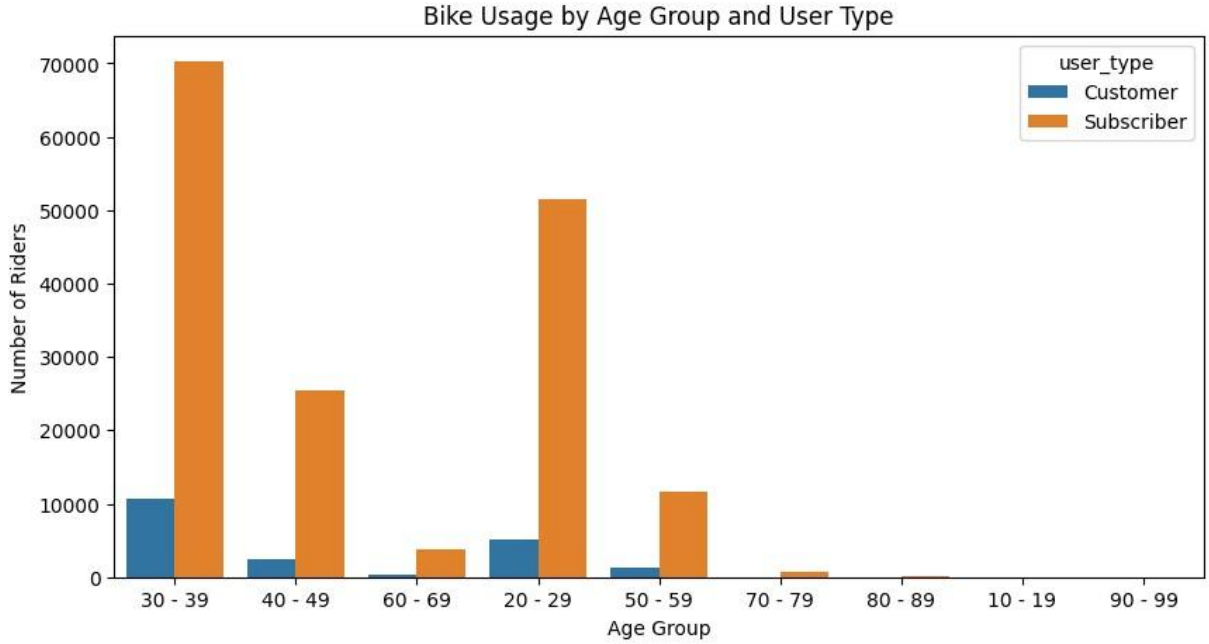
We also notice some outliers, especially users with ages over 100, which likely result from incorrect birth year entries. Cleaning the data to remove unrealistic ages and filtering extreme trip durations could help improve accuracy. Overall, this plot suggests that younger riders are more frequent and take longer trips, while older riders tend to keep their rides short.

How does trip duration vary between different user types (Subscribers vs. Customers)?



This visualization shows that Subscribers tend to take shorter and more consistent trips, likely for commuting, while Customers take longer trips with more variation, possibly for leisure or occasional use. Understanding this difference can help in optimizing pricing and marketing strategies for both user groups.

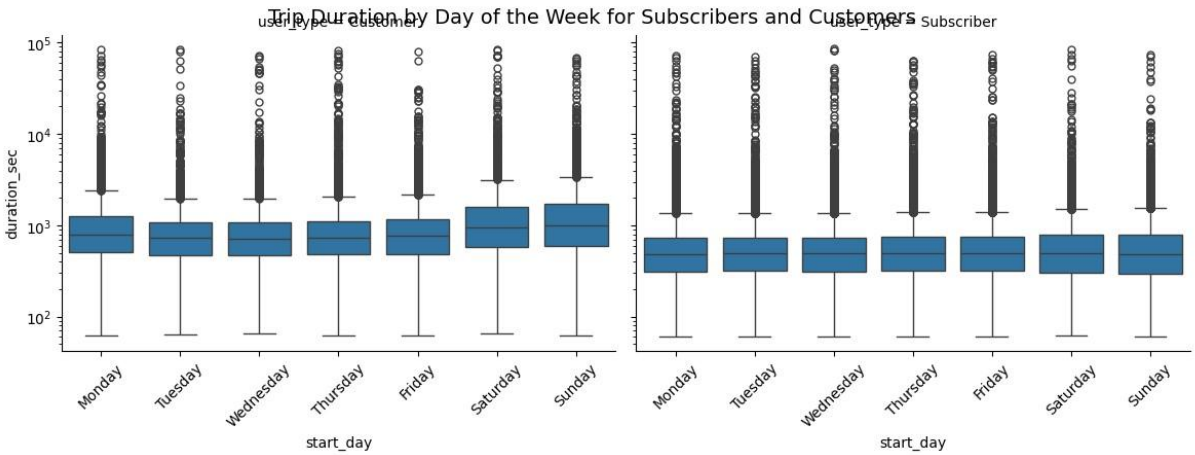
How does bike usage vary between different user types (Subscriber vs Customer) across age groups?



From this clustered bar chart, we can see how bike usage varies by age group and user type. The 30-39 age group has the highest number of riders, followed by 20-29. Across all age groups, subscribers significantly outnumber customers, suggesting that most riders prefer a membership-based system rather than occasional rentals. The difference is especially noticeable in younger age groups, where subscribers dominate. As age increases, the overall number of riders decreases, indicating that bike-sharing is more popular among younger adults.

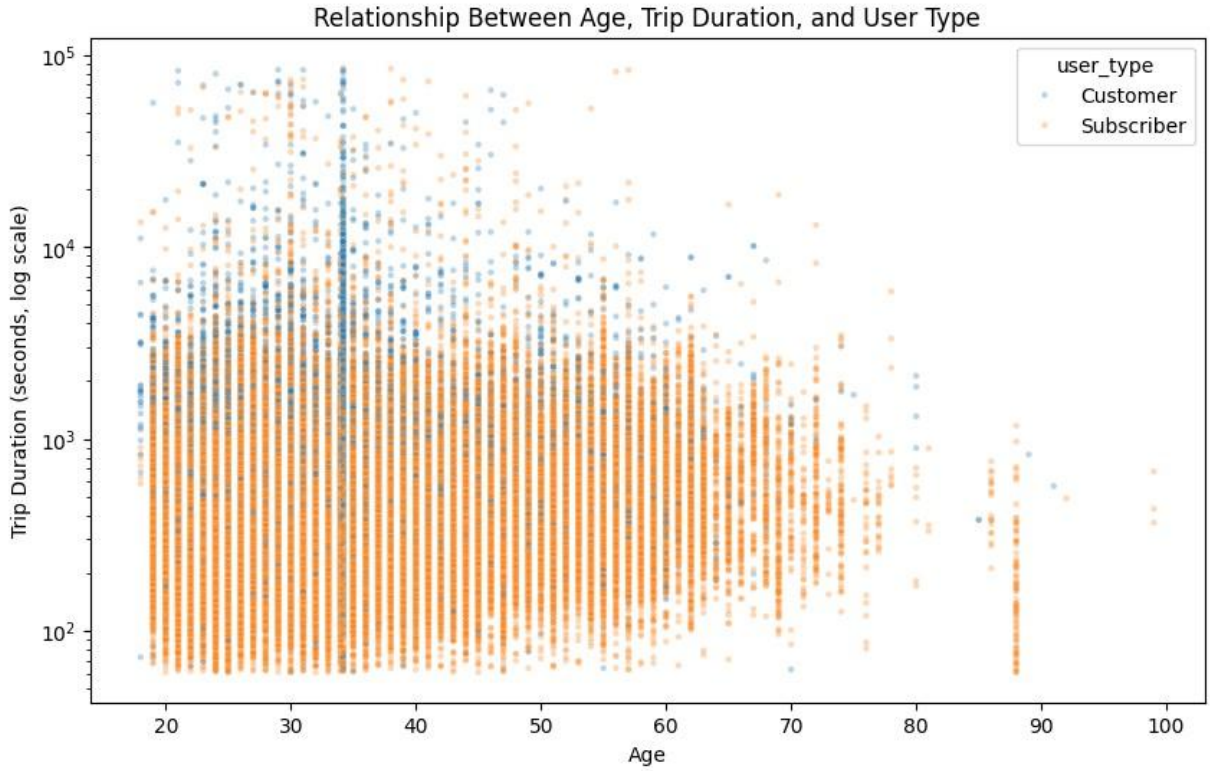
Multivariate Exploration

Do trip duration patterns differ by user type and day of the week?



From this faceted box plot, we can see how trip duration varies by day of the week for both customer and subscriber users. Customers generally have longer trip durations compared to subscribers, as shown by the higher median and wider spread of the box plots. This suggests that customers may use the service for leisure or infrequent trips, whereas subscribers tend to take shorter, more consistent trips, likely for commuting. Trip durations remain fairly consistent across all days of the week for both user types, indicating that weekday and weekend usage patterns are similar.

How do age, trip duration, and user type interact in the dataset? Are there noticeable trends based on user type?



From this scatter plot, we can observe the relationship between age, trip duration, and user type. Most trips are taken by users aged 20 to 40, with a noticeable decline in ridership as age increases. Across all age groups, subscribers outnumber customers, reinforcing the trend that most riders prefer a membership-based system.

Trip duration varies significantly, but most rides are concentrated within shorter durations, with a few outliers showing extremely long trips. Older riders (60+) participate less frequently, and their trips tend to be shorter in duration. The dense cluster at the bottom of the plot highlights that short-duration trips are the norm for all age groups, but customers appear to have slightly more variation in their trip times.

The data exploration is done !